Intro to Machine Learning
Quiz 4 Practice Problems
Practice problems, not turned in — Solutions

Directions: These practice problems cover the material for Quiz 4, which is an open-note in-class worksheet: cubic regression splines in Part 1 and smoothing splines in Part 2. They are not turned in, but the worksheet draws on them, so work through each one before looking at the solutions. You can bring them, with your solutions, to the worksheet. Problems adapted from ISL cite the exercise; see the book for the original wording.

Throughout, $h(x, \xi) = (x - \xi)^3_+$, which is $(x - \xi)^3$ for $x > \xi$ and 0 otherwise (ISL 7.10), and degrees of freedom (df) count every coefficient, including the intercept.

Part 1: Cubic Regression Splines

1. **The truncated power basis by hand.** (ISL 7.9, Exercise 1; ISL 7.4.3.)

   (a) Let $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 h(x, \xi)$. Show that f, f' and f'' are continuous at ξ, and find the jump in f''' at ξ.

   (b) A cubic spline with one knot at $\xi = 1$ was fitted, giving $\hat{f}(x) = 2 - x + x^2 + 3h(x, 1)$. Compute $\hat{f}(0)$ and $\hat{f}(2)$, and write $\hat{f}$ as a single cubic for $x > 1$.

   (c) A spline with knots at 0 and 1 is $\hat{f}(x) = x - h(x, 0) + 2h(x, 1)$. Compute $\hat{f}(-1)$, $\hat{f}(1)$ and $\hat{f}(2)$. At each knot, by how much does $\hat{f}'''$ jump?

   Solution.

   (a) Write $p(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$. Then $f(x) = p(x)$ for $x \le \xi$ and $f(x) = p(x) + \beta_4 (x - \xi)^3$ for $x > \xi$. The two pieces differ only by the extra term, and it and its first two derivatives, $\beta_4 (x - \xi)^3$, $3\beta_4 (x - \xi)^2$ and $6\beta_4 (x - \xi)$, are all 0 at $x = \xi$. So the two pieces agree at ξ in value, first derivative and second derivative: f, f' and f'' are continuous. The third derivative of the extra term is $6\beta_4$, so f''' jumps by $6\beta_4$ at ξ. Each truncated cubic adds exactly one df: one jump in the third derivative.

   (b) $\hat{f}(0) = 2$, since $h(0, 1) = 0$. $\hat{f}(2) = 2 - 2 + 4 + 3(1)^3 = 7$. For $x > 1$,

   $$\hat{f}(x) = 2 - x + x^2 + 3(x^3 - 3x^2 + 3x - 1) = -1 + 8x - 8x^2 + 3x^3,$$

   and as a check, $-1 + 16 - 32 + 24 = 7$.

   (c) $\hat{f}(-1) = -1$ (both truncated terms are 0). $\hat{f}(1) = 1 - 1^3 + 0 = 0$. $\hat{f}(2) = 2 - 2^3 + 2(1)^3 = -4$. By (a), $\hat{f}'''$ jumps by $6 \times (-1) = -6$ at $x = 0$ and by $6 \times 2 = 12$ at $x = 1$.

2. **Counting knots and degrees of freedom.** (ISL 7.4.1–7.4.3.)

   (a) You can afford df = 8. How many interior knots does a cubic spline get? A natural cubic spline?

   (b) With $K = 2$ knots, a piecewise cubic has $4 \times 3 = 12$ df. How many does it have if only f is required to be continuous at the knots? If f and f' are?

   (c) In one or two sentences, explain why a natural spline is usually preferred to a cubic spline with the same df.

   Solution.

   (a) Cubic spline: $K + 4 = 8$, so $K = 4$. Natural cubic spline: $K + 2 = 8$, so $K = 6$.

   (b) Each continuity condition at each knot removes one df. Only f continuous: $12 - 2 = 10$. f and f' continuous: $12 - 4 = 8$. (Adding f'' gives $12 - 6 = 6 = K + 4$, the cubic spline.)

   (c) The natural spline spends no df on the tails, where a cubic is fitted from few points and its variance is largest, and spends those df on extra knots inside the data. So it has lower variance at the edges for the same df (ISL Figure 7.4).

3. **Training RSS and nested fits.** (ISL 7.4.) True or false? Give a one-sentence reason for each.

   (a) Adding a knot to a cubic regression spline, and keeping the others, never increases the training RSS.

   (b) Adding a knot to a cubic regression spline can increase the test error.

   (c) Moving one knot to a new position, without changing the number of knots, never increases the training RSS.

   (d) With boundary knots at the smallest and largest x_i, a natural cubic spline has a training RSS at least as large as the cubic spline with the same interior knots.

   Solution.

   (a) True. The new basis function $h(x, \xi_{\text{new}})$ is added to the old ones, so the old fit is still available (coefficient 0 on the new one), and least squares can only do as well or better.

   (b) True. The extra flexibility lowers bias but raises variance, and past some point the variance wins (the U-curve).

   (c) False. The two fits use different basis functions, so neither model contains the other, and the new knot position can fit worse.

   (d) True. On $[x_1, x_n]$, the natural spline is a cubic spline with the same interior knots plus two extra constraints ($f'' = 0$ at both boundary knots), so it is a smaller model fitted to the same data. (Here the x_i are the data, not the knots.)

Part 2: Smoothing Splines

Here $\hat{g}_\lambda$ minimizes $\sum_{i=1}^{n}(y_i - g(x_i))^2 + \lambda \int g''(t)^2\,dt$ (ISL 7.11), $\mathbf{S}_\lambda$ is its smoother matrix, $\mathrm{df}_\lambda = \sum_i \{\mathbf{S}_\lambda\}_{ii}$ (ISL 7.13).

4. **Fit against roughness.** (ISL 7.5.1, eq. 7.11.) The x_i all lie in $[0, 1]$, and for this problem take the penalty integral over $[0, 1]$. Three candidate curves have these training RSS values:

Curve	$g(x)$	Training RSS
g_1	$1 + 2x$	2.0
g_2	$1 - x + x^2$	0.8
g_3	$1 - x + x^3$	0.2

   (a) Compute the penalty $\int_0^1 g''(t)^2\,dt$ for each curve.

   (b) Compute the criterion $\mathrm{RSS} + \lambda \int_0^1 g''(t)^2\,dt$ for each curve at $\lambda = 0.01$, 0.1 and 1. Which curve has the smallest criterion at each λ?

   (c) Describe in one sentence how the winner changes as λ grows, and why.

   (d) None of the three is the smoothing spline $\hat{g}_\lambda$. At $\lambda = 0.1$, what can you say about the criterion value of $\hat{g}_\lambda$ compared with your answers in (b)?

Solution.

(a) $g_1'' = 0$, so its penalty is 0. $g_2'' = 2$, so $\int_0^1 4\,dt = 4$. $g_3'' = 6t$, so $\int_0^1 36t^2\,dt = 12$.

(b)

λ	g_1	g_2	g_3
0.01	2.0	$0.8 + 0.04 = 0.84$	$0.2 + 0.12 = \mathbf{0.32}$
0.1	2.0	$0.8 + 0.4 = \mathbf{1.2}$	$0.2 + 1.2 = 1.4$
1	**2.0**	$0.8 + 4 = 4.8$	$0.2 + 12 = 12.2$

The smallest criterion (in bold) is g_3 at $\lambda = 0.01$, g_2 at $\lambda = 0.1$ and g_1 at $\lambda = 1$.

(c) As λ grows, roughness costs more, so the winner moves from the curve that fits best (g_3) to the smoothest one, the line g_1, whose penalty is 0 for every λ.

(d) It is at most 1.2. $\hat{g}_\lambda$ minimizes the criterion over *all* curves, g_1, g_2 and g_3 included, so its criterion is no larger than the best of them.

5. **Regression splines and smoothing splines side by side.** (ISL 7.4, 7.5.) Answer each in one or two sentences.

   (a) A cubic regression spline with K knots, fitted by least squares, has fitted values $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$. What is $\mathrm{tr}(\mathbf{H})$? What plays the role of $\mathbf{H}$ for a smoothing spline?

   (b) For a cubic regression spline you choose K and where the knots go. What do you choose for a smoothing spline?

   (c) Can a cubic regression spline have $\mathrm{df} = 6.4$? Can a smoothing spline?

   (d) A cubic regression spline with 2 knots and a smoothing spline with $\mathrm{df}_\lambda = 6$ are fitted to the same data. Both have 6 df. Name one way the two fits differ.

Solution.

(a) $\mathrm{tr}(\mathbf{H}) = K + 4$, the number of basis functions (the hat matrix of least squares has trace equal to the number of columns). For a smoothing spline the smoother matrix $\mathbf{S}_\lambda$ plays that role: $\hat{\mathbf{g}}_\lambda = \mathbf{S}_\lambda \mathbf{y}$ and $\mathrm{df}_\lambda = \mathrm{tr}(\mathbf{S}_\lambda)$.

(b) Only λ. The knots are fixed at every x_i, and λ (equivalently, df_λ) sets how much the coefficients are shrunk.

(c) The regression spline, no: its df counts basis functions, so it is a whole number. The smoothing spline, yes: df_λ is a sum of diagonal entries of $\mathbf{S}_\lambda$, which changes continuously with λ.

(d) Any one of these. The regression spline bends only at its 2 knots, while the smoothing spline can bend a little at every x_i. The regression spline is cubic beyond its boundary knots, while the smoothing spline is linear outside $[x_1, x_n]$. The regression spline's fit depends on where the 2 knots were put; the smoothing spline has no knots to place.

6. **What the smoothing spline is.** (ISL 7.5.1.) Answer each in one or two sentences.

(a) For $\lambda > 0$, what kind of function is $\hat{g}_\lambda$, and where are its knots?

(b) Suppose you took a natural cubic spline with a knot at every x_i and fitted it by ordinary least squares, with no penalty. What would the fitted curve do, and why? What stops the smoothing spline from doing the same?

(c) Nobody tells the smoothing spline to be linear beyond x_1 and x_n. Why is it?

(d) "A smoothing spline has n knots, so it has n parameters and must overfit." What is wrong with this?

Solution.

(a) A natural cubic spline with a knot at every (unique) x_i: cubic between consecutive x_i, with g, g' and g'' continuous, and linear outside $[x_1, x_n]$. Its coefficients are shrunk, by an amount that λ controls.

(b) It would pass through every data point. With a knot at every x_i, the natural spline has n coefficients for n points, so least squares can make every residual 0 (its basis matrix is square and invertible, so $\mathbf{H} = I$). The smoothing spline uses the same basis, but the penalty $\lambda \int g''(t)^2\, dt$ shrinks the coefficients, so for $\lambda > 0$ it does not interpolate (unless the points already lie on a line).

(c) There are no data beyond x_1 and x_n, so bending there adds to the penalty and does nothing for the RSS. The best curve doesn't bend there, so it is linear.

(d) The n coefficients are shrunk, so they are not free. Flexibility is measured by $\mathrm{df}_\lambda = \mathrm{tr}(\mathbf{S}_\lambda)$, which λ sets anywhere between 2 and n.

7. **The two ends of λ.** (ISL 7.5.1–7.5.2.) A smoothing spline is fitted to $n = 50$ points with distinct x_i.

(a) What are df_λ and the training RSS at $\lambda = 0$? What is $\mathbf{S}_0$?

(b) True or false: as $\lambda \to \infty$, $\hat{g}_\lambda$ tends to the horizontal line $\bar{y}$. Explain.

(c) Three fits have $\mathrm{df}_\lambda = 4$, 12 and 30. Which has the largest λ? Which has the largest training RSS?

(d) Can a smoothing spline have $\mathrm{df}_\lambda = 1.5$? Why or why not?

Solution.

(a) $\mathrm{df}_0 = 50$ and the training RSS is 0. With no penalty the fit interpolates, so $\hat{\mathbf{g}}_0 = \mathbf{y}$ and $\mathbf{S}_0 = I$.

(b) False. It tends to the least squares *line*. A line has $g'' = 0$, so its penalty is zero for every λ, and among lines the RSS picks the least squares one.

(c) df_λ decreases as λ increases, so $\mathrm{df} = 4$ has the largest λ. It is also the least flexible, so it has the largest training RSS.

(d) No. df_λ runs from n down to 2, never below: the intercept and slope are never penalized, so they always count fully.

8. **LOOCV without refitting.** (ISL 7.5.2 and 5.1.2, eq. 5.2.) For a smoothing spline,

$$\mathrm{RSS}_{cv}(\lambda) = \sum_{i=1}^{n} \left(y_i - \hat{g}_\lambda^{(-i)}(x_i)\right)^2 = \sum_{i=1}^{n} \left(\frac{y_i - \hat{g}_\lambda(x_i)}{1 - \{\mathbf{S}_\lambda\}_{ii}}\right)^2.$$

A smoothing spline is fitted to $n = 5$ points at two values of λ, with these residuals $e_i = y_i - \hat{g}_\lambda(x_i)$ and diagonal entries of $\mathbf{S}_\lambda$:

Fit		$i=1$	2	3	4	5
A	e_i	0.1	-0.2	0.25	-0.2	0.1
	$\{\mathbf{S}_\lambda\}_{ii}$	0.9	0.8	0.75	0.8	0.9
B	e_i	0.2	-0.3	0.35	-0.3	0.2
	$\{\mathbf{S}_\lambda\}_{ii}$	0.6	0.4	0.3	0.4	0.6

(a) Find df_λ for each fit. Which was fitted with the larger λ?

(b) Compute the training RSS and RSS_{cv} for each fit. Which λ does LOOCV choose?

(c) What does the left-hand side of the formula mean, and why is it the right thing to compare rather than the training RSS?

(d) Why can't the formula be used at $\lambda = 0$?

Solution.

(a) $\mathrm{df}_A = 0.9 + 0.8 + 0.75 + 0.8 + 0.9 = 4.15$ and $\mathrm{df}_B = 0.6 + 0.4 + 0.3 + 0.4 + 0.6 = 2.3$. B has the smaller df, so the larger λ.

(b) Training RSS: A, $0.01 + 0.04 + 0.0625 + 0.04 + 0.01 = 0.1625$; B, $0.04 + 0.09 + 0.1225 + 0.09 + 0.04 = 0.3825$. For RSS_{cv}, divide each residual by $1 - \{\mathbf{S}_\lambda\}_{ii}$:

$$A : \tfrac{0.1}{0.1}, \tfrac{-0.2}{0.2}, \tfrac{0.25}{0.25}, \cdots = (1, -1, 1, -1, 1), \quad \mathrm{RSS}_{cv} = 5;$$

$$B : \tfrac{0.2}{0.4}, \tfrac{-0.3}{0.6}, \tfrac{0.35}{0.7}, \cdots = (0.5, -0.5, 0.5, -0.5, 0.5), \quad \mathrm{RSS}_{cv} = 1.25.$$

LOOCV chooses B's λ, even though A has the smaller training RSS.

(c) $\hat{g}_\lambda^{(-i)}(x_i)$ is the prediction at x_i from the fit that never saw (x_i, y_i), so the left side measures prediction error on unseen points, an estimate of test error. The training RSS rewards fitting the noise: fit A nearly interpolates (its $\{\mathbf{S}_\lambda\}_{ii}$ are close to 1), so its residuals are small but its predictions of left-out points are poor.

(d) At $\lambda = 0$, $\mathbf{S}_0 = I$, so every $1 - \{\mathbf{S}_\lambda\}_{ii} = 0$ and every residual is 0: the formula is $0/0$. The fit interpolates, so the residuals carry no information about prediction error.